

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 7

DIVISION OF TWO 8-BIT NUMBERS WITH REMAINDER

AIM: To write and execute an 8085 assembly language program to divide two 8-bit numbers and store the result in memory.

ALGORITHM

1. Load the dividend into register A.
2. Load the divisor into register B.
3. Initialize register C to 00H.
4. Compare the dividend with the divisor.
5. If the dividend is smaller than the divisor, go to step 9.
6. Subtract the divisor from the dividend.
7. Increment register C.
8. Repeat steps 4–7.
9. Store the quotient from register C.
10. Store the remainder from register A.
11. Stop

CODE:

```
MVI A,14H
MVI B,04H
MVI C,00H
LOOP: CMP B
JC END
SUB B
INR C
JMP LOOP
END: STA 8051H
MOV A,C
STA 8050H
HLT
```

OUTPUT: